STATISTICS & MACHINE LEARNING · LESSON 22 OF 28

Clustering I: K-Means and RFM Customer Segmentation

Find groups without labels. Heuristic RFM segmentation (recency, frequency, monetary scores and named segments) as taught in class, then K-means clustering, scaling and log transforms, choosing k with the elbow method and silhouette score, profiling and naming clusters, and turning segments into marketing actions, on 4,500 retail transactions.

ML class 6 · ✦ scikit-learn 1.7+ · Intermediate · ⏱ 120 min

By the end of this lesson you can

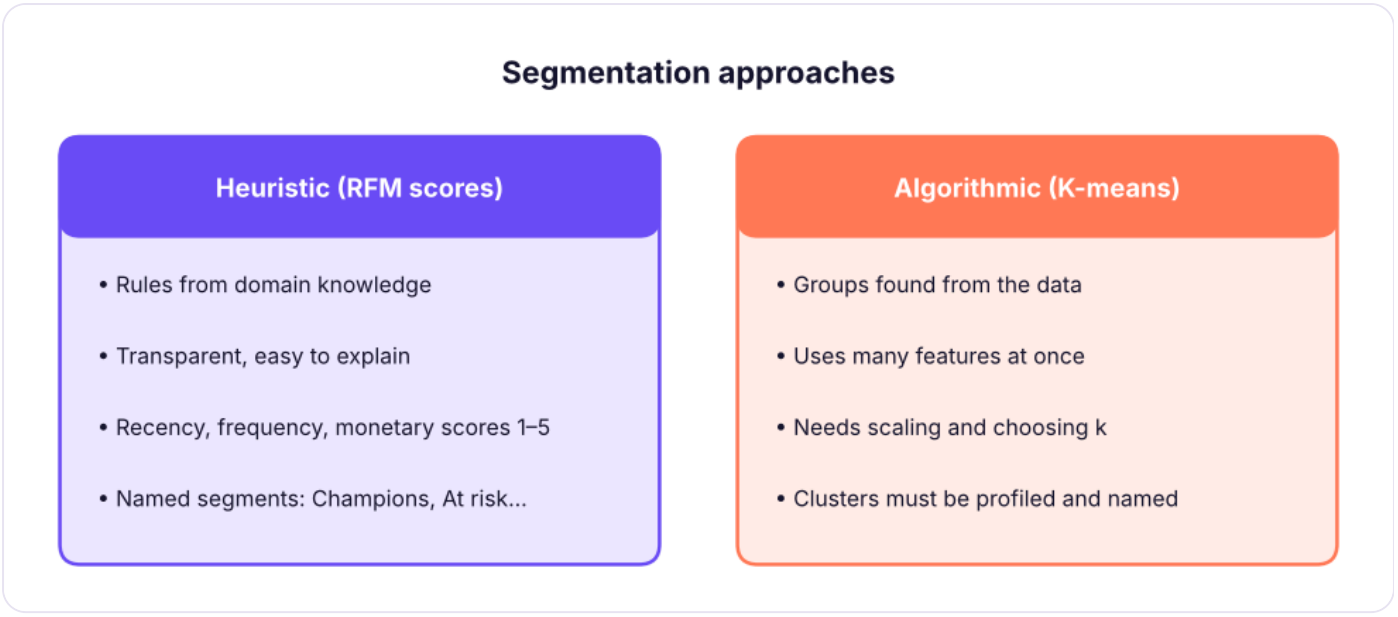
- Compute recency, frequency and monetary value per customer from transactions
- Build heuristic RFM scores and named segments (Champions, At risk, Hibernating...)
- Run K-means, explain its algorithm and its assumptions, and scale features properly
- Choose k with the elbow method and silhouette scores, and profile the clusters
- Translate segments into actions and measure them

Datasets used

`retail_transactions.csv`

Why segment customers?

Treating every customer the same wastes money: loyal customers get discounts they did not need, while slipping customers get nothing. **Segmentation** groups similar customers so each group gets the right message.



Two ways to segment: business rules (RFM) and algorithms (clustering).

Step 1: RFM table

Recency: days since the last purchase (lower is better). **Frequency**: number of purchases. **Monetary**: total spend.

Python

```
import pandas as pd, numpy as np
pd.set_option("display.max_columns", 20); pd.set_option("display.width", 160)
tx = pd.read_csv("retail_transactions.csv", parse_dates=["invoice_date"])
print(tx.shape, tx.customer_id.nunique(), "customers |", tx.invoice_date.min().date(),
"to", tx.invoice_date.max().date())
snapshot = tx.invoice_date.max() + pd.Timedelta(days=1)
rfm = tx.groupby("customer_id").agg(recency=("invoice_date", lambda d: (snapshot -
d.max()).days), frequency=("invoice_no", "nunique"), monetary=("amount", "sum"))
rfm["monetary"] = rfm.monetary.round(0)
print(rfm.describe().round(1).T[["mean", "50%", "min", "max"]])
print(rfm.head())
```

OUTPUT

```
(4548, 4) 500 customers | 2024-01-01 to 2025-12-22
              mean     50%    min      max
recency      217.0   144.0    1.0    722.0
frequency      9.1     6.0    1.0     39.0
monetary   11058.9  5122.0  196.0  76170.0
             recency  frequency  monetary
customer_id
R0001             53         33   54906.0
R0002            224          7    5426.0
```

```
R0003            26          33   56031.0
R0004           320           3    2017.0
R0005           109          35   49225.0
```

Step 2: heuristic RFM scores

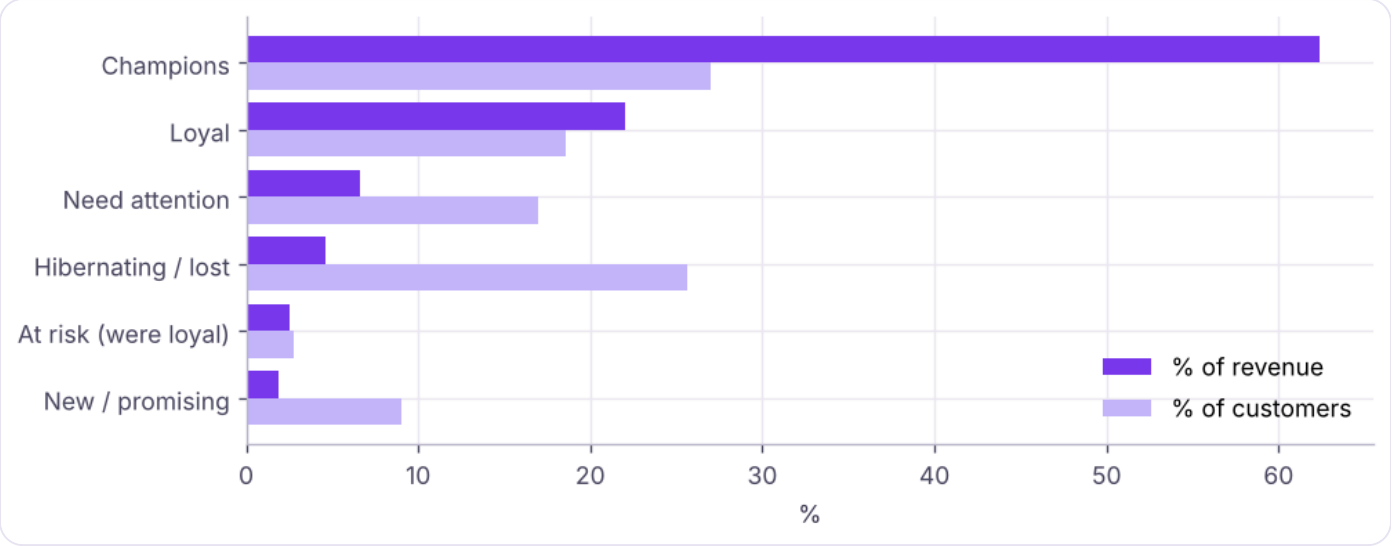
Score each dimension 1–5 by quintile (5 = best), then combine.

Python

```python
rfm["R"] = pd.qcut(rfm.recency, 5, labels=[5, 4, 3, 2, 1]).astype(int)           # recent = high score
rfm["F"] = pd.qcut(rfm.frequency.rank(method="first"), 5, labels=[1, 2, 3, 4, 5]).astype(int)
rfm["M"] = pd.qcut(rfm.monetary, 5, labels=[1, 2, 3, 4, 5]).astype(int)
rfm["RFM_code"] = rfm.R.astype(str) + rfm.F.astype(str) + rfm.M.astype(str)
rfm["RFM_score"] = rfm[["R", "F", "M"]].sum(axis=1)
def segment(r):
    if r.R ⩾ 4 and r.F ⩾ 4: return "Champions"
    if r.R ⩾ 3 and r.F ⩾ 3: return "Loyal"
    if r.R ⩾ 4 and r.F ⩽ 2: return "New / promising"
    if r.R ⩽ 2 and r.F ⩾ 4: return "At risk (were loyal)"
    if r.R ⩽ 2 and r.F ⩽ 2: return "Hibernating / lost"
    return "Need attention"
rfm["segment"] = rfm.apply(segment, axis=1)
seg = rfm.groupby("segment").agg(customers=("R", "size"), recency=("recency", "median"), frequency=("frequency", "median"), monetary=("monetary", "median"), revenue=("monetary", "sum"))
seg["revenue_%"] = (100 * seg.revenue / seg.revenue.sum()).round(1)
print(seg.sort_values("revenue", ascending=False).drop(columns="revenue"))
```

OUTPUT

segment	customers	recency	frequency	monetary	revenue_%
Champions	135	69.0	13.0	18414.0	62.4
Loyal	93	135.0	9.0	7844.0	22.0
Need attention	85	279.0	5.0	4236.0	6.6
Hibernating / lost	128	414.5	2.0	1836.5	4.6
At risk (were loyal)	14	235.5	9.0	7797.0	2.5
New / promising	45	80.0	3.0	2521.0	1.9



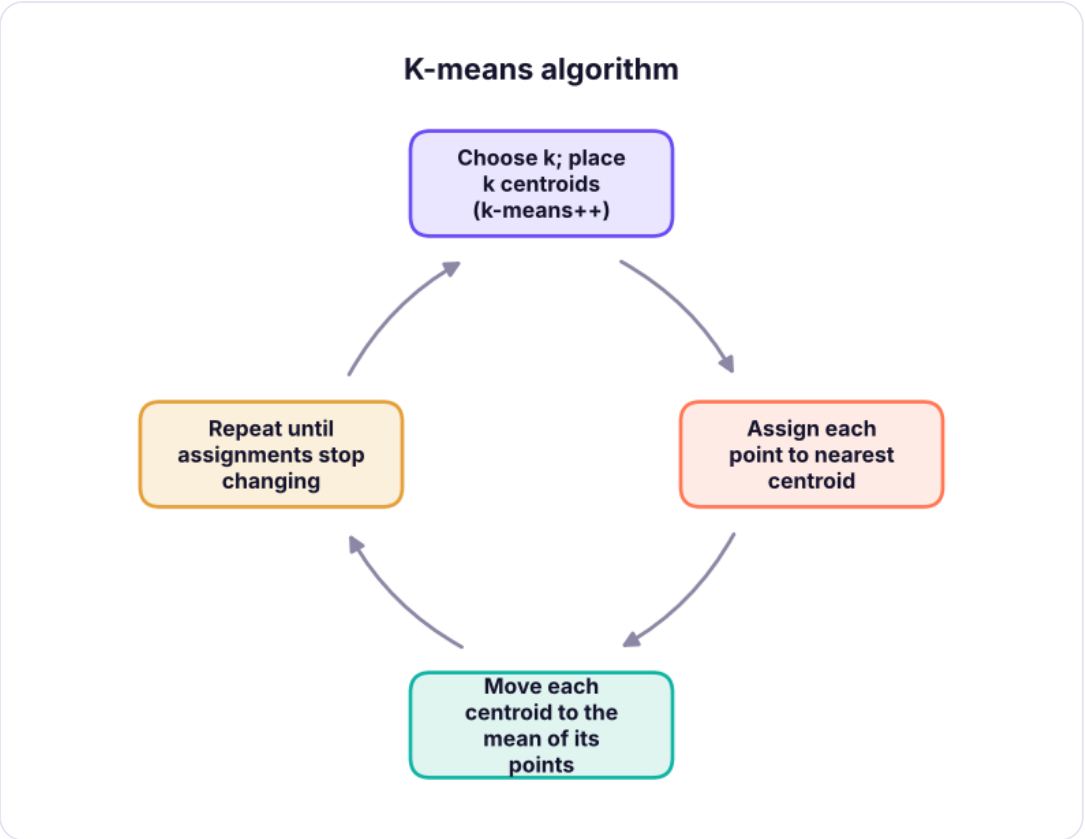
Revenue share vs customer share by RFM segment.

Segment	Typical action
Champions	Reward, early access, referrals; do not over-discount
Loyal	Upsell, loyalty points
New / promising	Onboarding, second-purchase offer
At risk (were loyal)	Win-back call or personalised offer **now**
Hibernating / lost	Low-cost reactivation; accept some loss
Need attention	Targeted reminders

RFM is transparent and works well, which is why the live class starts there. Its limits: fixed rules, only three variables, and quintile cut-offs that may not match natural groups.

Step 3: K-means clustering

K-means partitions data into *k* clusters so that each point belongs to the cluster with the nearest **centroid** (mean), minimising the within-cluster sum of squares (**inertia**).



The K-means loop: assign points to the nearest centroid, move centroids to the mean of their points, repeat until nothing changes.

Prepare features: log and scale

Python

```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
feat = pd.DataFrame({"log_recency": np.log1p(rfm.recency), "log_frequency":
np.log1p(rfm.frequency), "log_monetary": np.log1p(rfm.monetary)})
print("skew before log:", rfm[["recency", "frequency",
"monetary"]].skew().round(2).to_dict(), "| after:", feat.skew().round(2).to_dict())
Z = StandardScaler().fit_transform(feat)
```

OUTPUT

```
skew before log: {'recency': 1.01, 'frequency': 1.73, 'monetary': 2.09} | after:
{'log_recency': -0.71, 'log_frequency': 0.29, 'log_monetary': 0.12}
```

K-means uses Euclidean distance, so (1) **scale** features or the largest one (monetary) dominates, and (2) **log-transform** skewed variables so a few big spenders do not form their own clusters.

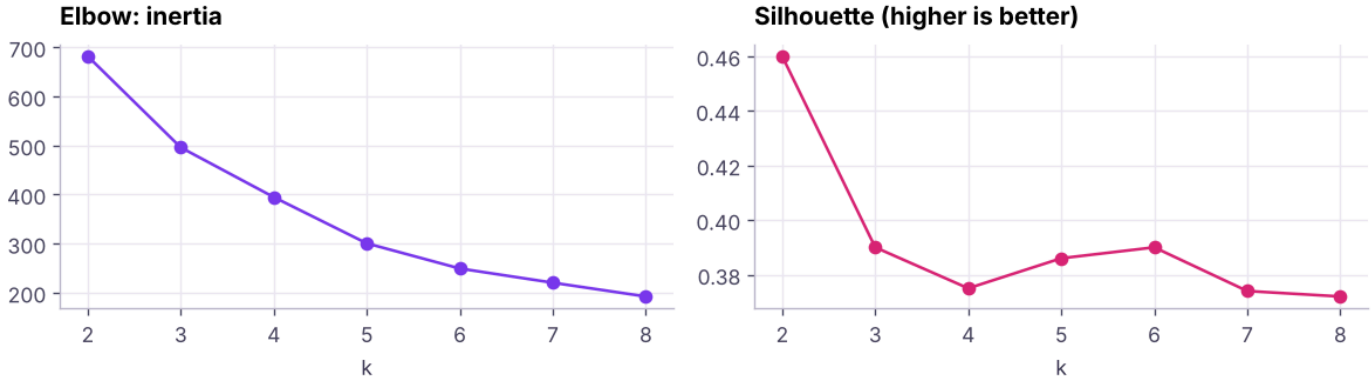
Choosing k

Python

```
from sklearn.metrics import silhouette_score, davies_bouldin_score
rows = []
for k in range(2, 9):
    km = KMeans(n_clusters=k, n_init=20, random_state=0).fit(Z)
    rows.append({"k": k, "inertia": round(km.inertia_, 1), "silhouette":
round(silhouette_score(Z, km.labels_), 3), "davies_bouldin": round(davies_bouldin_score(Z,
km.labels_), 3)})
kt = pd.DataFrame(rows); print(kt.to_string(index=False))
```

OUTPUT

k	inertia	silhouette	davies_bouldin
2	681.7	0.460	0.813
3	495.8	0.390	0.889
4	394.6	0.375	0.907
5	300.4	0.386	0.832
6	249.1	0.390	0.838
7	220.2	0.374	0.865
8	192.1	0.372	0.853



Elbow curve (inertia) and silhouette score by number of clusters.

- **Elbow method:** inertia always falls as k grows; look for the bend where extra clusters stop helping much.
- **Silhouette score** (−1 to 1): how close each point is to its own cluster compared with the next one; higher is better.
- **Davies-Bouldin** (lower is better).
- **Business sense:** the number of segments the team can actually act on (often 3–6).

Here the silhouette is highest at k = 2 (it simply splits active from lapsed customers), with no clear elbow afterwards. We choose **k = 4**: the scores are close, and four groups give the marketing team more useful distinctions. Statistics guide the choice; the business decides.

Fit and profile

Python

```
k = 4
km = KMeans(n_clusters=k, n_init=20, random_state=0).fit(Z)
rfm["cluster"] = km.labels_
prof = rfm.groupby("cluster").agg(customers=("R", "size"), recency=("recency", "median"),
frequency=("frequency", "median"), monetary=("monetary", "median"), revenue=("monetary",
"sum"))
prof["revenue_%"] = (100 * prof.revenue / prof.revenue.sum()).round(1)
prof = prof.sort_values("monetary", ascending=False); print(prof.drop(columns="revenue"))
```

OUTPUT

cluster	customers	recency	frequency	monetary	revenue_%
2	117	68.0	20.0	32923.0	69.3
0	164	165.5	8.0	6643.0	21.2
3	79	84.0	4.0	2836.0	4.5
1	140	406.0	2.0	1880.5	5.0

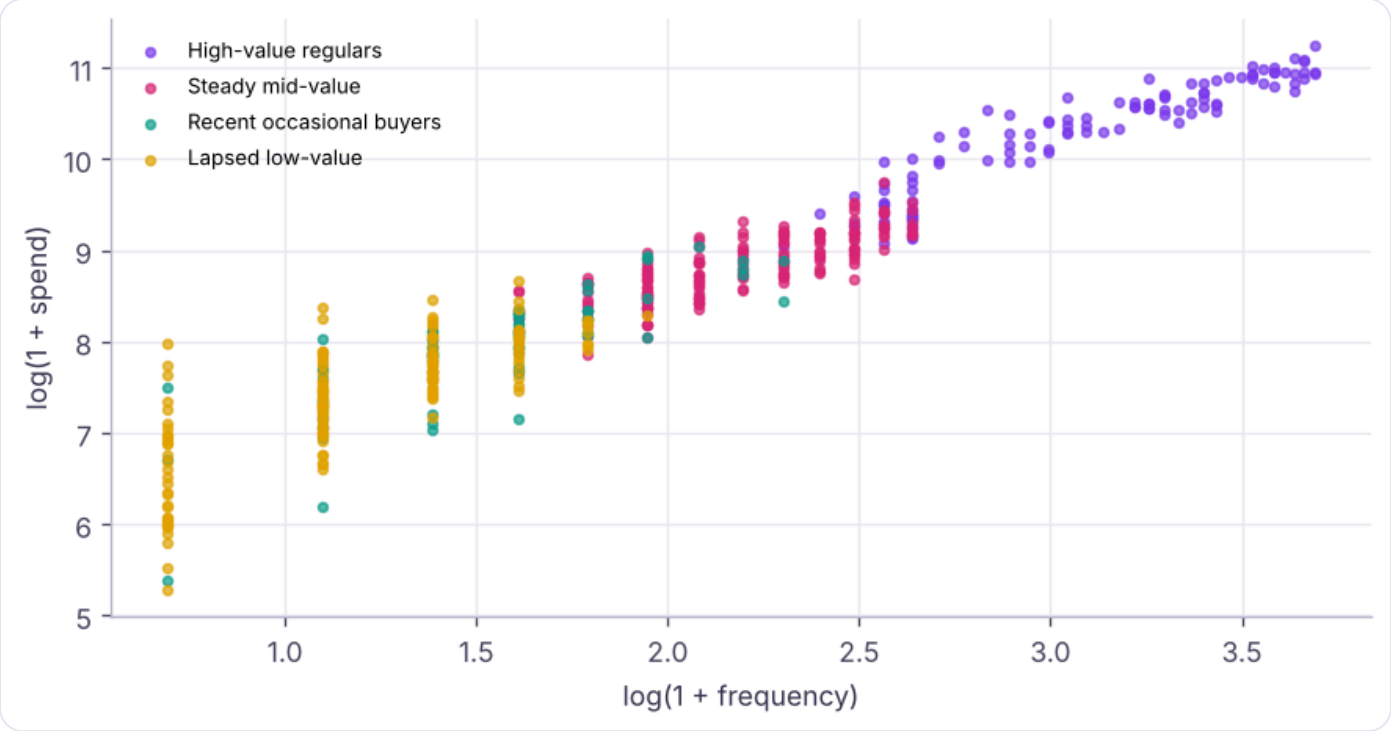
Python

```
order = prof.index.tolist()
names = dict(zip(order, ["High-value regulars", "Steady mid-value", "Recent occasional
buyers", "Lapsed low-value"]))
rfm["cluster_name"] = rfm.cluster.map(names)
print(pd.crosstab(rfm.cluster_name, rfm.segment))
```

OUTPUT

segment	At risk (were loyal)	Champions	Hibernating / lost	Loyal	Need attention	New / promising
cluster_name						
High-value regulars	1	101	0	15	0	0
Lapsed low-value	0	0	125	0	15	0
Recent occasional buyers	0	4	0	14	16	45
Steady mid-value	13	30	3	64	54	0

Name clusters from their **profiles** (median recency, frequency, spend), not from the cluster number. The cross-tab shows how the data-driven clusters line up with the rule-based RFM segments.



Customers by log frequency and log spend, coloured by K-means cluster (centroids marked).

K-means assumptions and limits

K-means assumes / needs	If not	Alternative
Roughly spherical, similar-size clusters	Elongated or uneven clusters split badly	Gaussian mixtures, DBSCAN (next lesson)
Numeric, scaled features	Categories distort distances	K-prototypes, Gower distance
k chosen in advance		Silhouette, business needs, hierarchical dendrogram
Sensitive to outliers and initialisation		Log/robust scaling, `n_init`, k-means++ (default)

From segments to action

1. Write a one-line profile and a goal for each segment (e.g. "At risk: bought often, silent for 4+ months → win back").
2. Design an offer or message per segment.
3. **Test** it: randomly hold out part of each segment as control (lesson 6) and measure incremental revenue.
4. Refresh the segmentation monthly and track how customers move between segments.

Summary

KEY TAKEAWAYS

- RFM turns transactions into recency, frequency and monetary value; quintile scores and rules give transparent named segments.
- K-means assigns points to the nearest of k centroids, minimising inertia; log-transform skewed variables and scale features first.
- Choose k with the elbow, silhouette and Davies-Bouldin scores plus business sense; profile and name clusters by their medians.
- K-means prefers spherical clusters and numeric data; use other methods for odd shapes or categorical data.
- Segments are only valuable when they drive tested actions.

Quick quiz

6 questions · tap an answer to check it

1. What does "recency" measure in RFM?

A Days since the customer's last purchase

B Number of purchases

C Total spend

D Days since first purchase

Answer: A. Lower recency = more recently active.

2. Why scale features before K-means?

A Distances would otherwise be dominated by the feature with the largest range

B K-means needs normal data

C To remove outliers

D It is optional

Answer: A. Spend in rupees would dwarf frequency.

3. What does inertia measure?

A Within-cluster sum of squared distances to centroids

B Distance between clusters

C Number of clusters

D Accuracy

Answer: A. It always decreases as k grows.

4. A silhouette score near 0 means...

A Points lie between clusters; clusters overlap

B Perfect clusters

C Wrong labels

D One cluster only

Answer: A. Closer to 1 is better separation.

5. A segment "bought often but not for 5 months" is best called...

A Champions

B At risk

C New customers

D Loyal

Answer: B. Action: win-back now.

6. How should you name clusters?

A From their profiles (median recency, frequency, spend and other traits)

B By cluster number

C Randomly

D By the largest customer

Answer: A. Names should describe behaviour.

EXERCISE

Practice

(1) Add `avg_order_value` and `tenure_days` (days since first purchase) to the features and re-cluster: does the silhouette improve? (2) Compare K-means with k = 3 and k = 5 profiles: which is more actionable? (3) Compute RFM on the Superstore data for 2025 and list the top 10 Champions.

Show solution (worked answer)

Python

```
first = tx.groupby("customer_id").invoice_date.min()
f2 = feat.assign(log_aov=np.log1p(rfm.monetary / rfm.frequency),
log_tenure=np.log1p((snapshot - first).dt.days))
Z2 = StandardScaler().fit_transform(f2)
for k in (3, 4, 5): print(k, silhouette_score(Z2, KMeans(k, n_init=20,
random_state=0).fit_predict(Z2)))
s = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"]); s =
s[s.order_date.dt.year == 2025]
snap = s.order_date.max() + pd.Timedelta(days=1)
r2 = s.groupby("customer_id").agg(recency=("order_date", lambda d: (snap -
d.max()).days), frequency=("order_id", "nunique"), monetary=("sales", "sum"))
r2["score"] = pd.qcut(-r2.recency, 5, labels=False) +
pd.qcut(r2.frequency.rank(method="first"), 5, labels=False) + pd.qcut(r2.monetary, 5,
labels=False)
print(r2.sort_values(["score", "monetary"], ascending=False).head(10))
```

INTERVIEW CORNER

Questions you may be asked

Q: How do you choose the number of clusters? Elbow plot of inertia, silhouette and Davies-Bouldin scores, stability across random seeds, and above all whether the segments are distinct and actionable.

Q: What are the limitations of K-means? k must be chosen; it assumes spherical, similar-sized clusters; it is sensitive to scaling, outliers and initialisation; it handles only numeric distances.

Q: Explain RFM segmentation. Score customers on recency, frequency and monetary value (usually by quintiles), combine the scores, and map combinations to named segments with tailored actions.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.